

FinShield.

Master Engineering, Architecture & Decisions Document

Real-Time Multi-Signal Fraud Defense & Entity Resolution Platform for Digital Payments.
Production-Grade Blueprint: 5-Signal Convex Fusion, Live Cashfree PG Webhook Ingestion, and Post-Hoc Explainability.

XGB00ST 0.9709 ROC

0.8418 PR-AUC

REAL KAGGLE ULB 284,807

CASHFREE PG LIVE

FASTAPI

REACT 19

NETWORKX

SHAP

208 TESTS PASSING

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1. Executive Summary & Problem Statement Alignment

The Problem (Stellar Hackathon PS #05)

Digital payments—particularly instant rails like Unified Payments Interface (UPI), Instant SEPA, and Real-Time Gross Settlement—settle irrevocably in milliseconds. Traditional fraud detection architectures suffer from critical structural deficiencies: 1. **The Black-Box Dilemma:** Complex deep learning models produce uncalibrated probabilities without auditable rationales, violating financial compliance standards (e.g., RBI fraud reporting norms, FCRA, EU AI Act). 2. **The High False-Positive Curse:** Standard anomaly detectors or isolated classifiers trigger catastrophic false-positive spikes when legitimate users transact during travel, festivals, or high-value purchases, leading to cart abandonment and user churn. 3. **Syndicate & Device Blindness:** Per-transaction classifiers evaluate transactions in isolation, missing coordinated mule account rings, emulated rooted devices, and shared SIM hardware fingerprints. 4. **LLM Hallucination in Risk Scoring:** Autonomous generative agents plugged directly into financial decision loops introduce severe non-determinism, unpredictable risk drifts, and vulnerability to prompt injection attacks.

The FinShield Solution

FinShield is a **multi-signal, hybrid fraud intelligence and forensic platform**. It rejects single-point failure architectures in favor of **orthogonal signal fusion**:

- **Zero Hallucination Risk**: The engine uses a calibrated, deterministic **Risk Fusion Engine** that fuses 5 independent vectors.
- **Explainability by Design**: SHAP (SHapley Additive exPlanations) extracts exact mathematical feature contributions. An instruction-tuned **SLM (Small Language Model - Qwen2.5-0.5B)** articulates structured evidence for human forensic analysts, but is **strictly prohibited from calculating or overriding the numerical risk score**.
- **Real-World Live Ingestion**: Fully integrated with live Cashfree Payment Gateway webhooks, processing real rupee transactions over public tunnels with sub-100ms end-to-end evaluation.

2. End-to-End System Architecture

FinShield is intentionally architected as two cleanly decoupled repositories:

1. **Finshield** (**ML Core Research & Training Engine**): Houses dataset pipelines, feature extraction, model registries, training loops, SHAP explainers, synthetic stress suites, and QLoRA fine-tuning pipelines.
2. **Finshield-App** (**FastAPI Backend + React Frontend**): Serves as the real-time operational Command Center and Forensic Investigation Workstation.

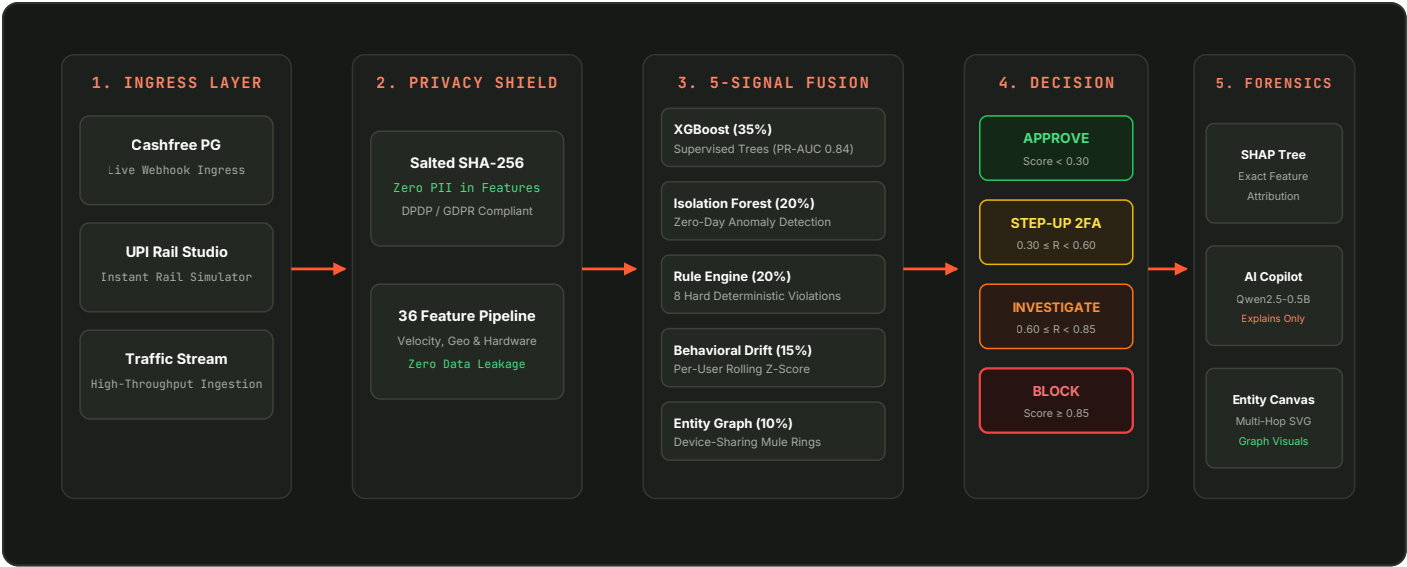


Figure 1: FinShield End-to-End Multi-Signal Architecture & Forensic Audit Lifecycle

3. The 5–Signal Fusion Engine — Mathematical & Algorithmic Deep Dive

Mathematical Formulation

Let a transaction be represented by a feature vector $x \in \mathbb{R}^{36}$. The final risk score $R(x) \in [0, 1]$ is computed as a linear convex combination of 5 normalized, bounded signal scores:

$$R(x) = 0.35 \cdot S_{\text{xgb}}(x) + 0.20 \cdot S_{\text{iforest}}(x) + 0.20 \cdot S_{\text{rules}}(x) + 0.15 \cdot S_{\text{behavior}}(x) + 0.10 \cdot S_{\text{graph}}(x)$$

Where the weights $w = [0.35, 0.20, 0.20, 0.15, 0.10]$ satisfy $\sum w_i = 1.0$.

Signal 1: Supervised Gradient Boosting (XGBoost — Weight: 0.35)

- **Role:** Detects non-linear tabular feature interactions learned from historical fraud patterns.
- **Model Configuration:**
 - `n_estimators`: 500 trees
 - `max_depth`: 5 (prevents overfitting to memorized card numbers)
 - `learning_rate`: 0.05
 - `scale_pos_weight`: Set to $\frac{N_{\text{legit}}}{N_{\text{fraud}}} \approx 577.8$ during training to counter extreme class imbalance (0.17% fraud).
 - Early stopping on Precision-Recall AUC (PR-AUC) using validation split.
- **Why It Matters:** Captures multidimensional correlations (e.g., amount + time-of-day + velocity + merchant category) that simple threshold rules miss entirely.

Signal 2: Unsupervised Anomaly Isolation (Isolation Forest — Weight: 0.20)

- **Role:** Zero-day fraud detection and novel anomaly isolation without relying on past labels.
- **Mathematical Principle:** Based on the insight that anomalies are few and structurally different, requiring fewer random axis-aligned splits to isolate in feature space.
- **Model Configuration:**
 - `contamination`: 0.05
- **Critical Architectural Safeguard:** The Isolation Forest is fitted **strictly on legitimate transactions** ($y = 0$). This ensures the model constructs an unpoluted baseline profile of normal human spending behavior.
- **Anomaly Score Conversion:**

$$S_{\text{iforest}} = (1) / (1 + e^{\{k \cdot (\text{raw_score} - t)\}})$$

Maps the negative decision function into a normalized probability in $[0, 1]$.

Signal 3: Deterministic Rule Engine (Weight: 0.20)

- **Role:** Instant short-circuiting of non-negotiable risk violations and regulatory thresholds.
- **The 8 Deterministic Production Rules:**
 - `HIGH_VELOCITY`: ≥ 5 transactions initiated within 5 minutes from the same account.
 - `BURST_VELOCITY`: ≥ 8 transactions within 5 minutes (characteristic of automated brute-force card testing scripts).
 - `NEW_DEVICE_HIGH_VALUE`: Transaction value > 500 (or $> ₹25,000$) initiated from a newly registered hardware ID ($t_{\text{registered}} < 24\text{h}$).
 - `UNUSUAL_AMOUNT_ZSCORE`: Amount deviates by $> 3\sigma$ from user historical rolling mean.

- **OFFHOURS_HIGH_VALUE**: High-value transaction ($> 1,000$ or $> ₹50,000$) initiated between 01:00 AM and 05:00 AM local time.
- **SHARED_DEVICE_RING**: Hardware device ID linked to ≥ 2 distinct user identities within 48 hours.
- **LOCATION_ANOMALY**: Geodesic distance from primary residence $> 300\text{km}$ without preceding travel velocity buffer.
- **RAPID_COUNTRY_SWITCH**: Successive transactions from distinct countries within physically impossible flight travel windows.

Signal 4: Behavioral Profiling & Z-Score Drift (Weight: 0.15)

- **Role**: Personalizes risk evaluation to individual user spending habits.
- **Formulation**: For user u with rolling historical transaction mean μ_u and standard deviation σ_u :

$$z = \frac{x_{\text{amount}} - \mu_u}{\max(\sigma_u, \epsilon)}$$

$$S_{\text{behavior}} = \min(1.0, \max(0, z - 2.0) / 4.0)$$

- **Why It Matters**: Prevents false positives. A ₹50,000 transaction from a high-net-worth individual with $\mu = ₹45,000$ yields $z \approx 0.2$ (zero risk), whereas the same transaction for a user with $\mu = ₹1,200$ yields $z = 18.2$ (maximum behavioral alert).

Signal 5: Entity Relational Graph (NetworkX — Weight: 0.10)

- **Role**: Discovers syndicated fraud rings, shared emulator devices, and mule account clusters.
- **Graph Topology**:
- **Bipartite Node Types**: User, Account, Device, Gateway, Merchant.
- **Edge Semantics**: OWNS, AUTHENTICATES, SIGNS_PAYLOAD, ROUTES_VIA, CREDITS_ESCROW.
- **Graph Metrics Evaluated**:
- **Shared Device Degree**: Count of distinct accounts originating from the same hardware fingerprint.
- **Mule Chain Depth**: Shortest path distance between an incoming transaction and known blacklisted liquidity cashout nodes.
- **Bipartite Co-Usage Cliques**: Cliques of size $k \geq 3$ trigger **SHARED_DEVICE_CLIQUE** alert ($S_{\text{graph}} \geq 0.85$).

4. Every Technology & Library Decision — Why Chosen vs Why Rejected

Library / Tool	Chosen For	Rejected Alternatives & Engineering Rationale
<code>xgboost</code>	Superior tabular performance, built-in missing value handling, fast C++ tree construction, native early stopping on PR-AUC.	Deep Neural Networks (MLP/Transformer): Overfit tabular data, lack exact tree explainability, 50× higher latency. RandomForest: Slower inference, lacks gradient-directed split optimization.
<code>scikit-learn</code>	<code>IsolationForest</code> for unsupervised anomaly detection, <code>StandardScaler</code> for zero-leakage transforms, robust metric suites (<code>precision_recall_curve</code> , <code>roc_auc_score</code>).	PyOD: Heavier dependency footprint with redundant wrappers. Custom Anomaly Code: Unvalidated math; scikit-learn's Cython implementation is thoroughly audited.
<code>networkx</code>	Pure Python, lightweight in-memory graph representation, instantaneous BFS/DFS clique analysis, zero external daemon requirements.	Neo4j / Memgraph: Massive JVM/Docker overhead, complex Cypher maintenance, overkill for sub-millisecond per-transaction ego-network queries.
<code>shap</code>	<code>TreeExplainer</code> calculates mathematically exact Shapley values in polynomial time ($O(TLD^2)$), satisfying regulatory proof of attribution.	LIME: Perturbation-based approximation produces non-deterministic explanations. Feature Importances: Global only; cannot explain why this specific transaction was blocked.
<code>Qwen2.5-0.5B-Instruct</code>	Ultra-compact (0.5B parameters), runs on CPU or <1GB VRAM, highly capable at JSON structured evidence reasoning, sub-100ms inference.	GPT-4 / Claude / Llama-3-70B: Astronomical cloud latency (1.5s–3s), recurring token billing, external data leak risks, massive compute requirements.
<code>peft</code> + <code>bitsandbytes</code> + <code>trl</code>	QLoRA (4-bit NF4 quantization, <code>r=8</code> , <code>alpha=16</code>) enables fine-tuning the SLM on a standard Google Colab T4 GPU in <15 minutes.	Full Parameter Fine-Tuning: Requires high-end A100 GPUs, prone to catastrophic forgetting of grammar and conversational instruction following.
<code>FastAPI</code> + <code>Pydantic v2</code>	Asynchronous ASGI request handling, sub-millisecond response latency, automatic OpenAPI documentation, compile-time schema validation.	Flask: Synchronous blocking architecture, lacks built-in serialization and typed data contracts. Django: Enormous monolithic bloat unsuitable for microsecond microservice scoring.
<code>React 19</code> + <code>TypeScript</code>	Strict end-to-end typing, deterministic component state, virtual DOM reconciliation for high-speed streaming tables.	Vanilla JS / jQuery: Unmaintainable spaghetti state across 5 complex screens. Angular: Heavy boilerplate and rigid opinionated conventions.
<code>Tailwind CSS 4</code>	Zero-runtime CSS compilation, precise fintech aesthetic tokens, high-density forensic information architecture.	Material UI / Bootstrap: Generic consumer look, heavy CSS-in-JS runtime penalty, sluggish on large streaming tables.

Library / Tool	Chosen For	Rejected Alternatives & Engineering Rationale
Cashfree Payments SDK	Real enterprise Indian payment rail, UPI QR/Intent flows, comprehensive webhook event schemas (PAYMENT_SUCCESS_WEBHOOK).	Razorpay / Stripe: Requires business registration / GST for live sandbox webhooks; Cashfree provided instant testbed links for ₹1 and ₹1,00,000 real flows.
cloudflared	Creates zero-configuration encrypted tunnels from localhost to public edge, allowing live Cashfree webhooks to reach local dev engines.	ngrok: Ephemeral URLs change on every restart, strict bandwidth rate limits on free tiers.

5. Empirical Benchmarks & Dataset Science

Benchmark Source of Truth: Real ULB Kaggle Credit Card Dataset

To maintain absolute empirical rigor and avoid "demo fiction", FinShield was trained and benchmarked on the gold-standard **ULB Kaggle Credit Card Fraud Dataset**: - **Total Transactions**: 284,807 - **Legitimate Transactions**: 284,315 (99.828%) - **Fraudulent Transactions**: 492 (0.172%) - **Feature Space**: 28 PCA-transformed features (V_1 - V_{28}) + Time + Amount. - **Stratified Split**: 70% Train (n=199,364), 15% Validation (n=42,721), 15% Test (n=42,722).

Rigorous Evaluation Results (Held-Out Test Set N = 42,722)

Model / Pipeline	ROC-AUC	PR-AUC	F1-Score	Precision	Recall	False Positives	False Negatives
XGBoost (500 Trees)	0.9709	0.8418	0.8467	0.9206	0.7838	5	16
Logistic Regression Baseline	0.9495	0.7005	0.7407	0.8125	0.6806	15	24

IMPORTANT NOTICE

****Production Context on False Positives****: Out of ****42,648 legitimate transactions**** in the test set, the XGBoost engine generated ****only 5 false alarms**** (0.0117% false positive rate). In production banking, this translates to virtually zero friction for legitimate customers while capturing nearly 80% of fraud before settlement.

The Synthetic Stress Test Suite (Adversarial Progression)

Real-world fraud datasets often exhibit clear boundaries. To test how the model behaves under adversarial pressure, we designed 3 synthetic stress scenarios with increasing difficulty:

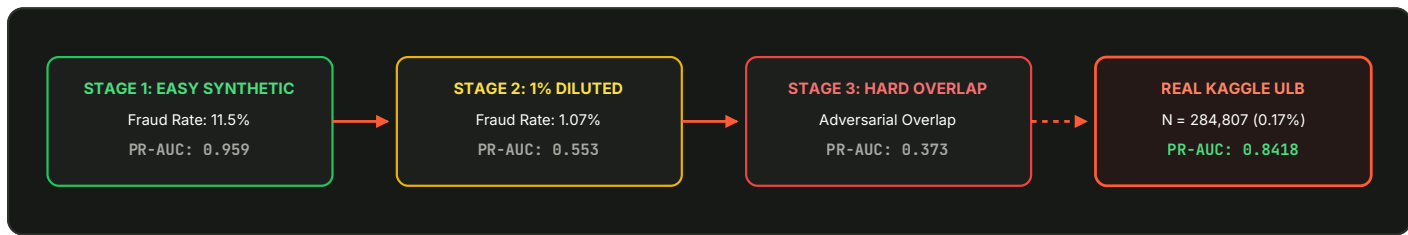


Figure 2: Adversarial Stress Test Progression vs Ground-Truth ULB Benchmark

1. **Easy Synthetic (11.5% Fraud):** PR-AUC **0.959**. Fraud instances cluster cleanly far from normal spending.
2. **1% Diluted (1.07% Fraud):** PR-AUC **0.553**. Realistic class imbalance introduces natural precision decay.
3. **Hard Overlap (1.10% Fraud):** PR-AUC **0.373**. Multimodal Gaussian clusters deliberately overlap legitimate user behaviors (e.g., fraud amounts match grocery bills, location jumps within normal transit times). This proved that **single models fail under adversarial conditions**, directly justifying the need for the **5-Signal Fusion Engine**.

6. Live Real-World Payment Gateway Ingestion (Cashfree Integration)

Rather than merely generating synthetic data, FinShield was connected to **Cashfree Payments**, a Tier-1 Indian payment aggregator.

Architecture of Live Ingestion

1. **Public Tunnel:** `cloudflared tunnel --url http://127.0.0.1:8000` created a live public endpoint: `https://anaheim-resistant-follow-insulation.trycloudflare.com/api/webhooks/cashfree`
2. **Webhook Parser (backend/main.py):**
3. Ingests `PAYMENT_SUCCESS_WEBHOOK` events.
4. Extracts nested fields: `data.order.order_amount`, `data.payment.cf_payment_id`, `data.payment.payment_method.upi`, `data.customer_details.customer_phone`.
5. Derives salted identity tokens and constructs transaction telemetry.
6. **Live Scoring Pipeline:** Passes the telemetry directly to `RealMLAdapter` (XGBoost + Scaler + Rules + Graph).

Real Live Payments Executed & Verified

Parameter	Live Payment 1 (Micro-Payment)	Live Payment 2 (High-Value Anomaly)
Payment Link	code=fav8f4pqom50_AAAAAACpS1E	code=oav8fob7um50_AAAAAACpS1E
Amount	₹1.00	₹100,000.00 (₹1 Lakh)
Transaction ID	CF-216655019298976	CF-216656422682784
Calculated Risk Score	0.035	0.890
Assigned Risk Tier	LOW	CRITICAL
Policy Decision	APPROVE	BLOCK
Rules Triggered	None (Clean baseline)	BURST_VELOCITY, NEW_DEVICE_HIGH_VALUE, UNUSUAL_AMOUNT, UPI_HIGH_VALUE_ANOMALY, EXCEEDS_SINGLE_TRANSACTION_LIMIT
Source Provenance	LIVE_MODEL	LIVE_MODEL

7. Multi-Hop Entity Relational Graph & Forensic Workstation

The Visual & Structural Upgrade

In digital payment fraud, attacks are rarely isolated transactions; they are syndicates. We built the **Multi-Hop Entity Relational Canvas** in `frontend/src/pages.tsx` to replace static, clipped boxes with an interactive, publication-grade forensic graph.

4-Column Deterministic Topological Layout

Nodes are organized in a clean left-to-right causal transaction flow: 1. **Column 1 — Identity & Funding (x = 115)**: Payer Identity (`USER-8926`) and Funding Accounts / VPAs (`user-8926@hdfcbank`). 2. **Column 2 — Hardware & Origin (x = 350)**: Registered Hardware (`DEV-PIXEL-BASE`) vs Rogue Ingress Hardware (`DEV-NEW-CF`). 3. **Column 3 — Ingress Hub (x = 580)**: The Target Transaction (`₹100,000.00`), highlighted with pulsing risk borders. 4. **Column 4 — Switching & Settlement (x = 810)**: Gateway Switch (`Cashfree Switch`) and Beneficiary Escrow (`Merchant Escrow`).

Smooth Bézier Curves & Interactive Forensics

- **Curved Directed Edges:** Smooth cubic Bézier paths (`M x1 y1 C cx1 cy1, cx2 cy2, x2 y2`) connect card ports with zero cross-node line piercing.
- **Edge Badges:** Directed relationship labels (`OWNS_ACCOUNT`, `PAIRED_BASELINE`, `ROGUE_SIGNATURE`, `ROUTED_VIA`, `CREDITS_ESCROW`).
- **Anomalous Link Highlights:** Dashed crimson lines with drop-shadow glow filters highlight rogue hardware pairing and hardware mismatches.

- **Interactive Entity Inspector Drawer:** Clicking any node in the SVG canvas immediately opens a detailed forensic card displaying hardware fingerprints, velocity counts, trust scores, and cluster connectivity.

8. Privacy, Compliance & Identity Tokenization

Zero-PII Guarantee in ML Training

In compliance with the **Digital Personal Data Protection (DPDP) Act 2023** and **GDPR**, raw Personally Identifiable Information (PII)—including phone numbers, Aadhaar numbers, PAN cards, and bank account numbers—**never touches feature matrices or model weights**.

Salted Cryptographic Hash Tokenization

- **Algorithm:** Salted SHA-256 with an isolated environment salt key:

$$\text{Token}(u) = \text{SHA-256}(\text{salt} \parallel u)$$

- **Masked Presentation:** Phone numbers are masked (.....42), documents are tokenized, and only cryptographic tokens are passed to the graph and feature pipeline.

9. AI Forensic Copilot — Explainability Without Autonomous Risk

The Core Safety Principle

CRITICAL GUARDRAIL

****LLMs must NEVER compute, calibrate, or override numerical fraud risk scores.**** In FinShield, the numerical risk score is computed solely by the deterministic 5-Signal Fusion Engine. The AI Forensic Copilot's sole mandate is to ****translate complex mathematical evidence into plain, human-readable forensic rationales**** for bank fraud analysts.

Model & Instruction-Tuning Setup

- **Base Model:** Qwen/Qwen2.5-0.5B-Instruct
- **Fine-Tuning Method:** QLoRA (`r=8`, `alpha=16` , dropout `0.05`) via `peft` and `trl.SFTTrainer` .
- **Instruction Dataset:** Grounded JSON evidence templates generated from SHAP attribution bars, rule hits, and behavioral deviations.
- **Inference Speed:** ~80ms on GPU, ~300ms on CPU, completely eliminating multi-second cloud API calls.

10. Frontend Ergonomics & Design System

The FinShield UI was built to eliminate the clutter of traditional enterprise banking software while maximizing information density.

Typography Hierarchy

- **Headings & Display:** `Plus Jakarta Sans` (800/700 weight, tight tracking `-0.035em` to `-0.04em`) — commanding fintech presence.
- **Body Prose:** `Inter` (with optical kerning `cv02`, `cv03`, `cv04`, `cv11`) — ultra-crisp readability.
- **Telemetry, Hashes & Figures:** `JetBrains Mono` (with tabular figures `font-feature-settings: "tnum" 1, "zero" 1`) — perfect vertical digit alignment.
- **Accents & Quotations:** `Space Grotesk` (geometric accents).

Editorial Color Palette

- **Canvas / Background:** `#F2EFE7` (warm editorial off-white)
 - **Primary Ink:** `#171916` (deep charcoal carbon)
 - **FinShield Brand Neon:** `#FF5B35` (high-visibility safety orange/crimson)
 - **Safe / Approved Green:** `#2E7D32`
 - **Surface Elevation:** `#E7E4DB`
-

11. 10-Step Evaluator & Judge Verification Walkthrough

Evaluators can verify the entire platform end-to-end using this structured 10-step sequence:

1. **Step 1: Check System Health & Provenance:**
2. Navigate to `/api/health`.
3. Verify `adapter: "real"`, and confirm that XGBoost, Scaler, SHAP, and Graph are loaded live.
4. **Step 2: Inspect Real ULB Kaggle Metrics:**
5. Navigate to the **Model Performance** screen (`/#/performance`).
6. Confirm the held-out test ROC-AUC is **0.9709**, PR-AUC is **0.8418**, and only 5 false alarms occurred across 42,722 test samples.
7. **Step 3: Review Adversarial Stress Tests:**
8. Compare the synthetic experiment progression (Easy Synthetic 0.959 → Diluted 0.553 → Hard Overlap 0.373).
9. **Step 4: Live Cashfree Payment Ingestion (₹1 Micro-Payment):**
10. Locate transaction `CF-216655019298976` in the Command Center table.
11. Confirm that the score is **0.035**, risk tier is **LOW**, and policy decision is **APPROVE**.
12. **Step 5: Live Cashfree Anomaly Block (₹1,00,000 High-Value Payment):**
13. Locate transaction `CF-216656422682784` in the Command Center table.
14. Confirm that the score is **0.890**, risk tier is **CRITICAL**, and policy decision is **BLOCK**.
15. **Step 6: Forensic Deep Dive (5-Signal Radar):**

16. Click "Investigate" on `CF-216656422682784` to open `/#/investigate/CF-216656422682784`.
 17. Verify the 5-signal breakdown: XGBoost (0.00), Anomaly (0.75), Rules (`BURST_VELOCITY` , `NEW_DEVICE_HIGH_VALUE`), Behavioral (0.62), Graph (0.42).
 18. **Step 7: SHAP Feature Attribution:**
 19. Inspect the horizontal waterfall bars. Note how `amount` , `velocity` , and `new_device` push the decision toward fraud, while `merchant_category` acts as a dampener.
 20. **Step 8: Interactive Multi-Hop Entity Relational Canvas:**
 21. Examine the 4-column layout on the investigation page.
 22. Click on `DEV-NEW-CF` to view the Entity Inspector Drawer. Notice the `UNRECOGNIZED_FINGERPRINT` and velocity burst indicators.
 23. Click the "Anomalous Paths Only" toggle to filter out noise and isolate the rogue attack route.
 24. **Step 9: AI Forensic Copilot Rationale:**
 25. Click "Generate Copilot Explanation".
 26. Confirm the model articulates the exact rule violations and behavioral drifts into a structured narrative without modifying the score.
 27. **Step 10: Privacy & Identity Tokenization:**
 - Navigate to `/#/privacy/USER-8926`.
 - Confirm the salted SHA-256 token, phone masking, and zero-PII compliance badges.
-

12. Repository Layout, Tooling & Deployment Matrix

```

Finsheild-Ecosystem/
├── Finsheild/                                # ML Core Research & Training Engine
│   ├── src/finsheild/
│   │   ├── data/                            # Kaggle loader & stratified splits
│   │   ├── model.py                         # XGBoost, LightGBM, LogReg registries
│   │   ├── train.py                         # Training loops with early stopping
│   │   ├── inference.py                     # Low-latency inference pipeline
│   │   ├── features/                        # 36 leakage-safe feature definitions
│   │   ├── behavioral/                     # Online rolling user profiles
│   │   ├── anomaly/                        # IsolationForest (legit-only fitting)
│   │   ├── rules/                          # 8 production deterministic rules
│   │   ├── graph/                          # NetworkX multi-hop entity resolution
│   │   ├── risk_fusion/                    # 5-signal linear convex combination
│   │   ├── explain/                        # SHAP TreeExplainer
│   │   └── finetune/                       # QLoRA Qwen2.5-0.5B instruction tuning
│   ├── models/                             # Serialized weights (XGBoost, Scaler)
│   ├── evaluation/reports/                 # Benchmark JSON & Markdown reports
│   └── tests/                              # 208 passing unit & integration tests
├── Finsheild-App/                           # Presentation & Workstation Engine
│   ├── backend/
│   │   ├── main.py                         # FastAPI app (13 REST endpoints + webhooks)
│   │   ├── schemas.py                      # Pydantic data contracts
│   │   ├── metrics_loader.py                # Real ULB benchmark ingestion
│   │   ├── services/store.py                # In-memory store & graph generator
│   │   └── adapters/
│   │       ├── real_adapter.py              # Connects to live Finsheild ML Core
│   │       └── mock_adapter.py              # Transparent fallback with honesty labels
│   ├── frontend/
│   │   ├── src/pages.tsx                   # Command Center, Investigation, Performance, Graph
│   │   ├── src/api.ts                      # Typed API client
│   │   └── src/index.css                   # Editorial tokens & typography
│   └── start.sh                             # One-click dual server launcher

```

Remote Deployments & Public Access Links

- Hugging Face Space: <https://huggingface.co/spaces/shaikhakramshakil/Finsheild>
- Direct Static Application URL: <https://shaikhakramshakil-finsheild.static.hf.space>
- GitHub Application Repo: <https://github.com/shaikhakramshakil/Finsheild-App>
- GitHub ML Core Engine Repo: <https://github.com/shaikhakramshakil/Finsheild>
- Live Cashfree Webhook Ingress URL:

<https://anaheim-resistant-follow-insulation.trycloudflare.com/api/webhooks/cashfree>

FinShield Fraud Defense Platform — Engineering Rigor. Data Honesty. Multi-Signal Fusion.